

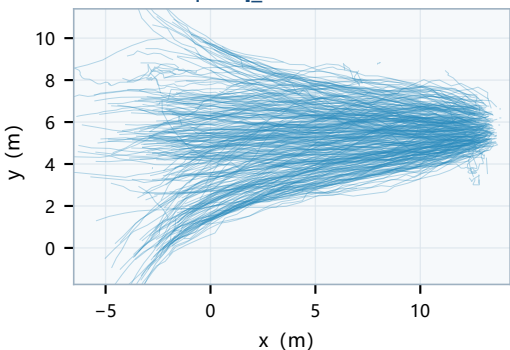
ETH-UCY 行人交通流特征分析

五个自然场景 · 统一 2.5 Hz 轨迹采样 · 从个体运动到群体交通流

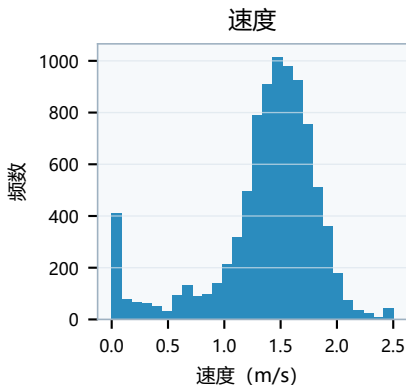
核心发现

拥挤度升高 → 邻距缩短、速度降低、交互增多

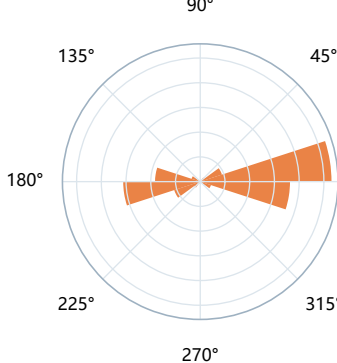
A1 轨迹图 | seq_eth



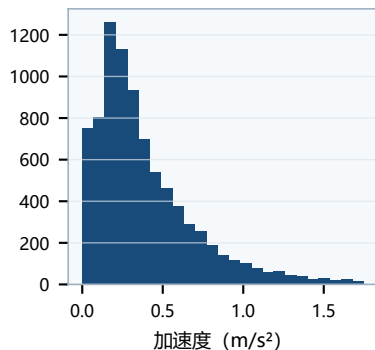
A2 个体运动学 | seq_eth



方向

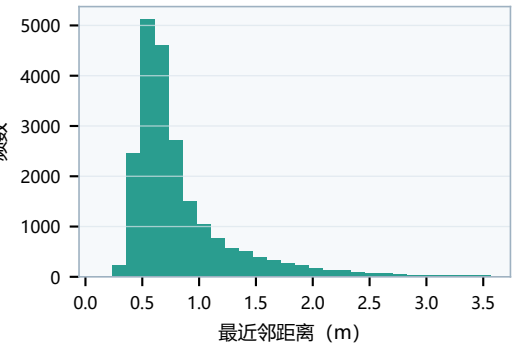


加速度

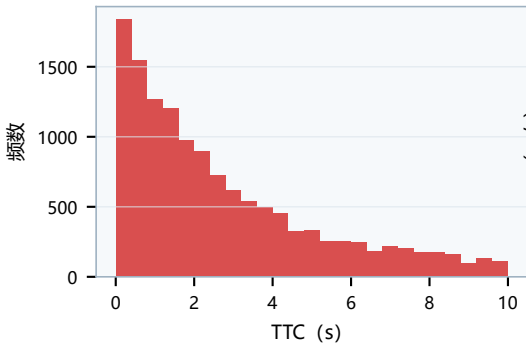


A3 最近邻与 TTC | students03

间距

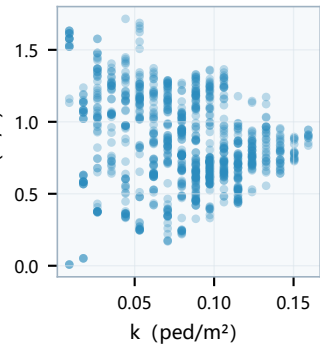


碰撞时间

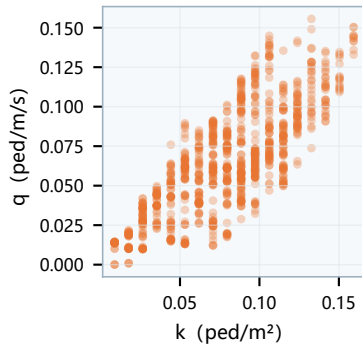


A4 q-k-v 基本图 | zara02

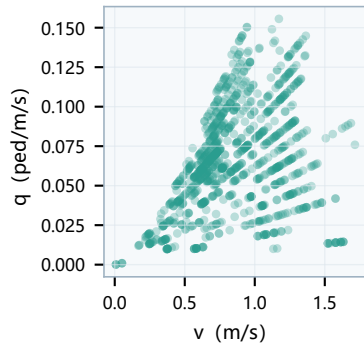
k-v



q-k

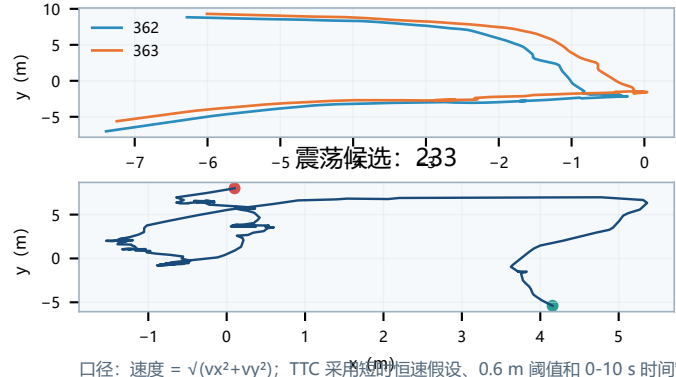


q-v

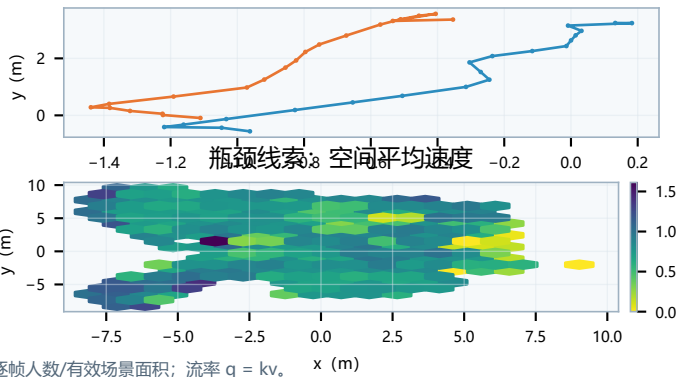


A5 行为分析 | students03

成行: 362-363



避让: 232-238



五场景对比与结论

场景	人数	速度	邻距
seq_eth	360	1.38	1.10
seq_hotel	390	1.04	1.19
zara01	148	1.11	0.97
zara02	204	0.82	0.80
students03	428	0.62	0.69

A1 seq_eth: 横向主流清晰, 侧向人流汇入。
A2 seq_eth: 均速 1.38 m/s, 运动整体稳定。
A3 students03: 邻距最小, 有限 TTC 事件最多。
A4 zara02: 密度升高时平均速度总体下降。
A5 students03: 成行、避让与局部低速并存。

行为图为规则筛选候选, 需结合视频人工复核。